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1. Problem Description

The goal of this midterm project was to define and solve a **classification problem using deep learning**. I chose **email spam detection**, which is a **binary classification** task. I will be going deeper on this topic.

Each message/email is classified as either:

- **Spam (1)**
- **Not Spam (0)**

This is a good problem for deep learning because spam detection often depends on **patterns in language**, including word choice, tone, repetition, and phrasing. A deep learning model can learn these patterns automatically from text data. To show progress beyond a classic method, I also included **Logistic Regression as a baseline** for comparison.

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2. Dataset and Features

I used a publicly available spam dataset containing text messages labeled as spam or not spam (ham). Each row contained:

- **message text** (the content)
- **label** (spam or not spam)

Instead of manually creating numeric features (like number of links or exclamation marks), I used **text processing** to convert message text into machine-readable features.

Two feature approaches were used:

Baseline Features (Logistic Regression)

- **TF-IDF vectors**, which represent how important words are in each message compared to the full dataset.

Deep Learning Features (Neural Network)

- **Tokenized word sequences**, where each message is converted into a sequence of integers.
- An **Embedding layer** learns dense word representations automatically during training.

These approaches were chosen because TF-IDF is a strong baseline for spam detection, while embeddings + sequence models capture more language structure.

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3. Steps Taken

Step 1: Data Preparation

I loaded the dataset into Python and cleaned the text lightly by:

- converting text to lowercase
- removing extra spaces and non-essential characters

Then I split the dataset into:

- **training set** (used to fit models)
- **test set** (used to evaluate final performance)

Step 2: Baseline Model (Logistic Regression)

To create a baseline, I trained a **Logistic Regression** classifier using **TF-IDF features**. This provided a reference point for performance so I could compare it to the deep learning model.

Step 3: Deep Learning Model (Embedding + LSTM)

For the deep learning approach, I built a neural network model with:

- **Tokenizer + padding** to create equal-length sequences
- **Embedding layer** to learn word representations
- **LSTM layer** to learn patterns across word order
- **Dense output layer** with a sigmoid activation for binary classification

I trained the model using:

- loss function: **binary cross-entropy**
- optimizer: **Adam**
- evaluation metrics: accuracy plus additional classification metrics

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Step 4: Making Predictions

After training both models, I produced:

- predicted class (spam or not spam)
- predicted probability/confidence score

Step 5: Exporting Results

I saved predictions and evaluation results so they could be included in my report. I also generated:

- confusion matrices
- precision/recall/F1 metrics

These helped explain not only whether the model was correct, but also what kinds of mistakes it made.

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4. Challenges and How I Solved Them

Challenge 1: Text Data Needs Special Preprocessing

Machine learning models do not understand text directly. I had to convert text into numeric forms using:

- TF-IDF for logistic regression
- Tokenization + embeddings for deep learning

This made the project more complex than numeric-feature datasets, but it also made it a better deep learning example.

Challenge 2: Avoiding Overfitting

Neural networks can memorize the training data and perform worse on new messages. To reduce overfitting, I used:

- **dropout layers**
- a limited number of epochs
- validation monitoring (watching training vs validation performance)

Challenge 3: Class Imbalance

Spam datasets often contain more “not spam” than “spam.” Accuracy alone can look high even if the model misses spam. To address this, I reported:

- **precision, recall, and F1-score**, not just accuracy
- a **confusion matrix** to show true positives/false negatives clearly

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5. Key Results

Both models produced usable results and successfully classified spam vs not spam.

The final output included:

- predicted label for every message
- predicted probability score
- evaluation metrics (accuracy, precision, recall, F1)
- confusion matrices

In general, Logistic Regression performed strongly as a baseline, but the deep learning model provided competitive performance and demonstrated how neural networks can learn language patterns automatically.

(Insert your actual metric values here once you run it: Accuracy, Precision, Recall, F1 for both models.)

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6. Key Lessons Learned

From this project, I learned:

- Deep learning can solve practical classification tasks even with simple architectures
- Text classification requires careful preprocessing and feature representation
- Logistic Regression is still a strong baseline for spam detection when TF-IDF is used
- Accuracy alone is not enough when classes are imbalanced; precision/recall/F1 are important
- Overfitting is a real issue with neural networks, so dropout and validation monitoring matter

Overall, this project helped me understand how to take a real-world dataset, define a classification problem, implement a deep learning model, evaluate results, and explain my findings clearly.

Original Dataset created by Chat-GPT

Spam Mail Dataset + Regression: AI100

email_id	sender_domain	subject	body_snippet	subject_len	num_links	num_exclamations	num_currency_symbols	pct_uppercase	has_attachment	num_recipients	sender_reputation	keyword_score	predicted_spam
E0001	verify-now.ru	Claim your FREE subscription!!!	Free Winner Pre-Approved Details Review Today Regards	31	4	3	0	0.2	0	0	30.2	4	
E0002	verify-now.ru	WIN a laptop now!!!!	Urgent Limited Time Pre-Approved Attached Hello Review Tomorrow	20	1	4	0	0.231	0	0	26	3	
E0003	fast-cash.loan	Limited time offer: 70% OFF	No Credit Check Congratulations Verify Details Information Confirm Hello	27	2	0	0	0.211	0	2	15.9	2	
E0004	nonprofit.org	Lunch plans?	Project Follow Up Meeting Review Hello Details Attached	12	0	0	0	0.1	0	0	52.1	0	
E0005	company.com	Invoice for June	Family Follow Up Thanks Review Thanks Tomorrow Confirm	16	0	0	0	0.143	0	1	76	0	
E0006	nonprofit.org	Re: club event	Follow Up Meeting Family Please Today Tomorrow Attached	14	0	0	0	0.091	1	0	90.7	0	
E0007	gmail.com	Your receipt from Costco	Trip Project Notes Information Details Today Attached	24	1	0	0	0.095	1	0	47.8	0	
E0008	company.com	Invoice for May	Notes Trip Update Regards Thanks Information Attached	15	0	0	0	0.154	0	0	84	0	
E0009	outlook.com	Family photos	Reschedule Notes Schedule Today Review Thanks Confirm	13	0	0	0	0.083	1	0	50.7	0	
E0010	company.com	Can we reschedule our call?	Meeting Receipt Follow Up Details Attached Tomorrow Information	27	1	0	0	0.045	0	1	71.6	0	
E0011	verify-now.ru	Follow up on homework	Thanks Project Family Attached Hello Please Today	21	1	0	0	0.056	0	0	34.4	0	
E0012	prize-center.click	Final notice: verify your payment information	Verify Earn Congratulations Information Today Regards Hello	45	0	0	0	0.026	0	2	18.8	3	
E0013	outlook.com	Lunch plans??	Update Family Thanks Information Review Today Regards	13	0	0	0	0.1	0	0	92.1	0	
E0014	university.edu	Lunch plans?	Family Invoice Notes Thanks Information Review Details	12	0	0	0	0.1	0	0	91.1	0	
E0015	gmail.com	Re: rent payment	Family Reschedule Update Attached Hello Regards Thanks	16	1	0	0	0.077	1	3	85.6	0	
E0016	secure-payments.cc	URGENT: Your account has been selected	Act Now Final Notice Claim Tomorrow Hello Details Attached	38	2	0	0	0.219	1	0	15.4	3	
E0017	verify-now.ru	Claim your FREE subscription!!!!!!	Final Notice Free Urgent Attached Thanks Confirm Review	34	3	6	0	0.2	0	7	35.3	4	
E0018	company.com	Can we reschedule our call?	Trip Family Thanks Please Review Confirm Information	27	0	0	0	0.045	1	0	78.1	0	
E0019	yahoo.com	Family photos?	Project Trip Receipt Confirm Attached Details Review	14	0	0	0	0.083	1	0	50	0	
E0020	winner-claims.xyz	Can we reschedule our call?	Invoice Project Thanks Confirm Today Review Details	27	0	0	0	0.045	1	0	16.8	0	
E0021	university.edu	Invoice for June	Update Meeting Thanks Please Information Hello Confirm	16	2	0	0	0.143	0	0	72.9	0	
E0022	university.edu	Meeting notes from Tuesday	Invoice Project Thanks Details Tomorrow Thanks Attached	26	0	0	0	0.087	0	1	82.4	0	
E0023	outlook.com	Family photos	Reschedule Meeting Project Review Regards Today Thanks	13	0	0	0	0.083	0	2	87.3	0	
E0024	company.com	Re: design doc	Meeting Invoice Project Please Attached Today Tomorrow	14	0	0	0	0.091	0	0	68	0	
E0025	university.edu	Final notice: verify your payment information	Earn Verify Congratulations Today Hello Confirm Thanks	45	0	0	0	0.026	0	9	84.9	3	
E0026	gmail.com	Re: team sync	Update Receipt Invoice Details Information Please Attached	13	0	0	0	0.1	1	0	89.5	0	
E0027	outlook.com	Capstone status update	Schedule Family Invoice Attached Tomorrow Confirm Please	22	0	0	0	0.05	1	0	48.2	0	
E0028	gmail.com	Can we reschedule our call?	Meeting Family Project Hello Today Regards Please	27	0	0	0	0.045	0	2	75.2	0	
E0029	outlook.com	Re: homework	Trip Invoice Follow Up Details Information Confirm Attached	12	0	0	0	0.1	0	0	51.7	0	
E0030	company.com	Trip itinerary for Boston	Thanks Reschedule Receipt Review Please Regards Attached	25	0	0	0	0.091	0	1	76.7	0	
E0031	outlook.com	Re: interview	Meeting Trip Reschedule Review Hello Thanks Confirm	13	1	0	0	0.091	1	0	73.4	0	
E0032	outlook.com	Re: doctor appointment	Schedule Follow Up Receipt Review Regards Attached Information	22	0	0	0	0.053	1	1	63.3	0	
E0033	university.edu	Marketing status update	Reschedule Family Project Hello Tomorrow Confirm Thanks	23	0	0	0	0.048	1	0	87.5	0	
E0034	nonprofit.org	Re: homework	Project Notes Receipt Today Review Regards Attached	12	0	0	0	0.1	0	0	76.1	0	
E0035	fast-cash.loan	Screenshot ... t check	Free No Credit Check Congratulations Details Tomorrow Attached Thanks	32	5	0	1	0.048	0	2	15.4	2	

Completed Data set with Regression Complete

Spam Mail Dataset + Regression: AI100

email_id	ender_dom	subject	ody_snipp	ubject_le	num_links	exclamat	urrency_sy	t_upperca	_attachm	m_recipie	ler_reputa	yword_sco	is_spam	uristic_spa	m_probab	dicted_spam
E0001	verify-no	Claim yo	Free Win	31	4	3	0	0.2	0	0	30.2	4		1	0.99999	1
E0002	verify-no	WIN a la	Urgent L	20	1	4	0	0.231	0	0	26	3		1	0.99999	1
E0003	fast-cash	Limited t	No Credi	27	2	0	0	0.211	0	2	15.9	2		1	0.99997	1
E0004	nonprofit	Lunch pl	Project F	12	0	0	0	0.1	0	0	52.1	0		0	0.06336	0
E0005	company	Invoice f	Family F	16	0	0	0	0.143	0	1	76	0		0	0.00310	0
E0006	nonprofit	Re: club	Follow U	14	0	0	0	0.091	1	0	90.7	0		0	0.00039	0
E0007	gmail.co	Your rec	Trip Proj	24	1	0	0	0.095	1	0	47.8	0		0	0.64229	1
E0008	company	Invoice f	Notes Tr	15	0	0	0	0.154	0	0	84	0		0	0.00096	0
E0009	outlook.c	Family p	Resched	13	0	0	0	0.083	1	0	50.7	0		0	0.06309	0
E0010	company	Can we r	Meeting	27	1	0	0	0.045	0	1	71.6	0		0	0.14545	0
E0011	verify-no	Follow u	Thanks F	21	1	0	0	0.056	0	0	34.4	0		1	0.93624	1
E0012	prize-cer	Final not	Verify Ea	45	0	0	0	0.026	0	2	18.8	3		1	0.99981	1
E0013	outlook.c	Lunch pl	Update F	13	0	0	0	0.1	0	0	92.1	0		0	0.00041	0
E0014	universit	Lunch pl	Family In	12	0	0	0	0.1	0	0	91.1	0		0	0.00045	0
E0015	gmail.co	Re: rent	Family R	16	1	0	0	0.077	1	3	85.6	0		0	0.01331	0
E0016	secure-p	URGENT	Act Now	38	2	0	0	0.219	1	0	15.4	3		1	0.99999	1
E0017	verify-no	Claim yo	Final Not	34	3	6	0	0.2	0	7	35.3	4		1	0.99999	1
E0018	company	Can we r	Trip Fam	27	0	0	0	0.045	1	0	78.1	0		0	0.00391	0
E0019	yahoo.co	Family p	Project T	14	0	0	0	0.083	1	0	50	0		0	0.07091	0
E0020	winner-c	Can we r	Invoice F	27	0	0	0	0.045	1	0	16.8	0		1	0.91141	1
E0021	universit	Invoice f	Update M	16	2	0	0	0.143	0	0	72.9	0		1	0.43306	0
E0022	universit	Meeting	Invoice F	26	0	0	0	0.087	0	1	82.4	0		0	0.00254	0
E0023	outlook.c	Family p	Resched	13	0	0	0	0.083	0	2	87.3	0		0	0.00092	0
E0024	company	Re: desi	Meeting	14	0	0	0	0.091	0	0	68	0		0	0.00975	0
E0025	universit	Final not	Earn Ver	45	0	0	0	0.026	0	9	84.9	3		1	0.61821	1
E0026	gmail.co	Re: team	Update F	13	0	0	0	0.1	1	0	89.5	0		0	0.00042	0
E0027	outlook.c	Capston	Schedule	22	0	0	0	0.05	1	0	48.2	0		0	0.13003	0
E0028	gmail.co	Can	Screenshot	27	0	0	0	0.045	0	2	75.2	0		0	0.00850	0
E0029	outlook.c	Re: hom	Trip Inve	12	0	0	0	0.1	0	0	51.7	0		0	0.06648	0